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IoT Based Patient Health Monitoring System

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I. INTRODUCTION

In recent years, the integration of the Internet of Things (IoT) into healthcare systems has transformed the way patient data is collected, monitored, and analyzed. Traditional patient monitoring methods require continuous manual supervision and frequent hospital visits, which can be time-consuming, labor-intensive, and prone to human error. This is particularly challenging for elderly individuals, chronically ill patients, or those in remote areas who need regular health assessments but lack easy access to medical facilities. The advancement of IoT technologies, along with the availability of low-cost sensors and cloud-based analytics, provides an innovative approach to remote health monitoring by enabling real-time data collection and analysis of vital health parameters.

This project presents the design and development of an IoT-Based Patient Health Monitoring System that continuously tracks key physiological parameters such as heart rate, body temperature, and electrocardiogram (ECG) signals. The system utilizes a NodeMCU microcontroller, which serves as the central processing unit for data collection and transmission. Equipped with a built-in ESP8266 Wi-Fi module, the NodeMCU facilitates seamless wireless connectivity, allowing the acquired sensor data to be sent to the ThingSpeak cloud platform for real-time visualization, storage, and analysis. This integration enables both healthcare professionals and patients' families to remotely monitor vital health metrics from anywhere, at any time, using a web interface or mobile device.

The heart rate sensor measures the patient's pulse through optical sensing, while the temperature sensor captures body temperature variations. The ECG sensor records the electrical activity of the heart, providing insights into cardiac rhythm and possible irregularities. The data collected from these sensors are digitized and processed by the NodeMCU before being transmitted to the cloud. The ThingSpeak platform, through its powerful analytical tools and visualization capabilities, displays real-time graphs and charts for each parameter. In addition, threshold-based alerts can be configured to notify doctors or caregivers instantly if any abnormal readings are detected, ensuring prompt medical attention and intervention.

The system's architecture combines hardware sensing, cloud connectivity, and data analytics, forming a robust and scalable health monitoring framework. Moreover, the incorporation of machine learning algorithms enhances the system's intelligence by analyzing trends and predicting

possible anomalies in the patient's health patterns. Such predictive capabilities can help in early diagnosis and prevention of critical health events, thereby reducing the burden on hospitals and improving patient outcomes.

This project addresses several key challenges in traditional healthcare systems, such as limited monitoring frequency, geographical barriers, and delayed diagnosis. By leveraging IoT, it ensures that patient data is continuously tracked and accessible in real time. The proposed system is cost-effective, energy-efficient, and easy to deploy, making it suitable for home-based monitoring, rural healthcare setups, and telemedicine applications.

In summary, the IoT-Based Patient Health Monitoring System aims to revolutionize the healthcare domain by providing a reliable, real-time, and remote monitoring solution. It not only enhances the quality of patient care but also contributes to global healthcare goals by improving accessibility, promoting preventive health practices, and reducing the need for in-person consultations. As an IoT-driven innovation, this project exemplifies how emerging technologies can support the United Nations Sustainable Development Goal (SDG 3) – “Good Health and Well-Being” – by ensuring healthy lives and promoting well-being for all at all ages.

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II. LITERATURE SURVEY

Healthcare monitoring has evolved significantly with the advent of the Internet of Things (IoT). The integration of IoT with biomedical sensors enables real-time monitoring of vital health parameters, thereby improving diagnostic accuracy, reducing manual intervention, and ensuring timely medical care. Several research studies have contributed to the development of IoT-based health monitoring systems, focusing on various physiological parameters, communication modules, and data visualization platforms.

IoT-Based Health Monitoring Systems

The increasing demand for continuous health supervision has led researchers to develop IoT-based systems capable of monitoring patients remotely.

Ufoaroh et al. (2024) proposed an IoT-based Patient Health Monitoring System using Arduino Uno and the ESP8266 Wi-Fi module integrated with the BMP085 sensor. The system was designed to measure heart rate, body temperature, and blood pressure in real time. A major highlight of their work was the implementation of an alert mechanism that automatically sent SMS notifications to caregivers when any physiological parameter exceeded safe thresholds. This research demonstrated that IoT technology can enhance healthcare accessibility and ensure rapid response during emergencies [1].

IoT with ThingSpeak Platform

Paul Stone Brown Macheso and Angel G. Meela (2021) developed an IoT-based Patient Health Monitoring System using ESP8266 and Arduino, connected to the ThingSpeak cloud platform. Their system measured pulse rate (in BPM) and body temperature, allowing both local display on an LCD and remote visualization through the internet. The use of the ThingSpeak IoT platform made it possible to monitor patients' conditions from any location, thereby reducing the dependency on in-person consultations. This project emphasized cost-effectiveness, ease of access, and scalability as major advantages of IoT-enabled healthcare systems [2].

Comparative Analysis of Existing Systems

The reviewed studies reveal that IoT has been widely adopted for healthcare applications due to its affordability, reliability, and real-time data capabilities. While systems like those developed by Ufoaroh et al. [1] focused on SMS-based alerting, others like Macheso and Meela [2] utilized advanced cloud visualization for improved accessibility. Similarly, Bhardwaj et al. [3] highlighted

automation and physician alert systems to enhance responsiveness during emergencies.

However, most of the existing models are limited to monitoring a small set of physiological parameters and lack integrated analytics for predictive health assessment. Furthermore, data security, energy efficiency, and scalability remain significant challenges that must be addressed to make such systems viable for large-scale deployment.

Sensor Technologies for Vital-Sign Monitoring:

A range of sensing modalities is commonly used in IoT patient monitoring:

Heart-rate / Pulse sensors (PPG): Optical photoplethysmography (PPG) sensors are widely used for noninvasive pulse measurement. They offer simplicity, low cost, and reasonable accuracy at rest; however, motion artifacts can degrade performance and must be mitigated in hardware or software.

Body temperature sensors: Digital temperature sensors provide continuous body temperature recordings which, when combined with other vitals, aid in fever detection and infection monitoring.

Cloud Platforms and Visualization (ThingSpeak, Others):

Cloud platforms designed for IoT (ThingSpeak, Blynk, Adafruit IO, AWS IoT) offer rapid deployment of dashboards, data storage, basic analytics, and alerting. ThingSpeak, specifically, is frequently used in academic projects for:

- Time-series storage of sensor channels,
- Visualization widgets (line plots, gauges),
- Basic Matlab-based analytics and threshold-based alerts.
- Many published student and research projects use ThingSpeak for initial demonstrations because it integrates easily with Arduino/ESP toolchains and provides simple API keys for secure posting

Sensors Used in Health Monitoring Systems

Sensors are the key components in IoT-based healthcare systems, responsible for collecting real-time physiological data that reflect a patient's health condition. Different types of sensors have been integrated into IoT frameworks for accurate monitoring and diagnosis.

The heart rate sensor, based on photoplethysmography (PPG), measures the variation in blood flow to determine pulse rate. It provides a non-invasive and efficient means of monitoring cardiovascular activity and has been effectively implemented in multiple IoT-based studies [1], [2].

The temperature sensor records the body temperature of patients and is crucial for detecting fever or infection. Digital sensors such as DHT11 is widely used due to their accuracy, low cost, and ease of integration with microcontrollers like Arduino and NodeMCU [1].

The ECG sensor captures the electrical activity of the heart, enabling early detection of cardiac abnormalities. Research by Bhardwaj et al. (2022) demonstrated the significance of ECG sensors in IoT-enabled frameworks for continuous cardiac health evaluation [3].

Together, these sensors enable multi-parameter monitoring and facilitate real-time data transmission to cloud platforms. The integration of heart rate, temperature, and ECG sensors in the current project provides a comprehensive and reliable health monitoring approach for both clinical and home-based environments.

ThingSpeak Platform for IoT Data Visualization

ThingSpeak is a cloud-based platform widely used in IoT applications for real-time data storage, visualization, and analysis. It provides an Application Programming Interface (API) that allows microcontrollers such as NodeMCU and ESP8266 to transmit sensor data over the internet. The collected data can be displayed using graphical dashboards and monitored remotely through web or mobile interfaces.

Macheso and Meela (2021) utilized ThingSpeak in their IoT-based patient health monitoring system to visualize pulse rate and body temperature data. Their work demonstrated the efficiency of the platform in enabling remote healthcare supervision through real-time graph updates and alert mechanisms [2]. Similarly, ThingSpeak's integration with MATLAB analytics allows for threshold detection and automated alerts when sensor values exceed normal limits.

In this project, the ThingSpeak platform serves as the primary medium for cloud connectivity, data logging, and visualization, ensuring continuous monitoring and accessibility of patient health data to doctors and caregivers.

Machine Learning in Health Monitoring Systems

Machine learning (ML) has become an essential component in modern IoT-based healthcare systems, enhancing their ability to detect anomalies and predict potential health risks. Unlike conventional systems that rely solely on threshold values, ML algorithms analyze patterns and trends within physiological data to identify irregularities more accurately.

Recent studies have applied supervised learning techniques such as Support Vector Machines (SVM) and Decision Trees to classify health conditions based on heart rate, ECG, and temperature readings. Bhardwaj et al. (2022) incorporated data analytics within their IoT framework to alert physicians proactively for rapid diagnosis during the COVID-19 pandemic [3]. Such approaches improve system reliability by reducing false alarms and enabling early intervention.

In the present system, ML algorithms are employed on the ThingSpeak platform to analyze patient data trends, detect abnormal health conditions, and predict possible risks. This integration makes the system more intelligent, responsive, and suitable for continuous remote monitoring applications.

III. METHODOLOGY

This section explains the design, hardware implementation, software configuration, and operational flow of the IoT-based Patient Health Monitoring System. The objective is to create a real-time, low-cost system capable of continuously monitoring vital health parameters such as heart rate, body temperature, and ECG, and transmitting the data to the ThingSpeak cloud platform for visualization and analysis.

1. Hardware Components and Setup

- **NodeMCU Microcontroller:** The NodeMCU serves as the brain of the system. It is built around the ESP8266 Wi-Fi SoC, which provides sufficient processing power, analog and digital I/O pins, and integrated wireless capability. It reads data from the connected sensors, converts it into digital format using its ADC (Analog-to-Digital Converter), and transmits the data to the ThingSpeak server over Wi-Fi..

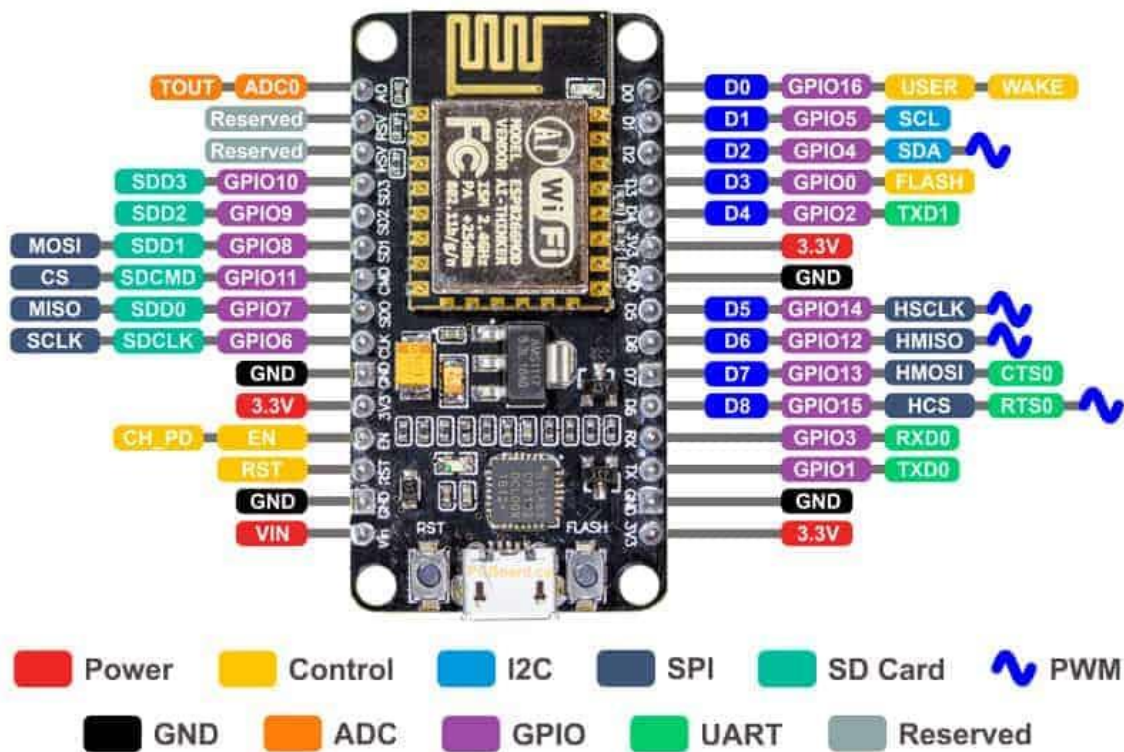


Fig 1. Detailed pinout diagram for an ESP8266 NodeMCU Wi-Fi development board

- **Heart Rate Sensor:** The heart rate sensor measures the patient's pulse using the principle of photoplethysmography (PPG), detecting blood flow changes to calculate beats per minute (BPM). It provides accurate, non-invasive monitoring of cardiac activity in real time.

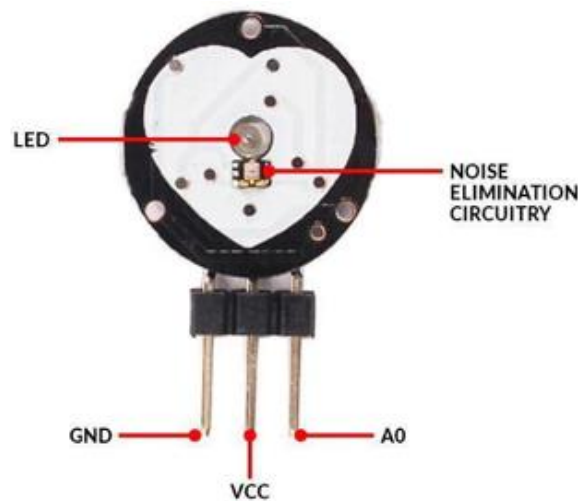


Fig 2. Pinout diagram for a heart-rate pulse sensor module

- **ECG Sensor (AD8232 module):** The ECG sensor captures the heart's electrical signals and generates an ECG waveform. This enables the detection of irregular heart rhythms and supports continuous cardiac health monitoring.

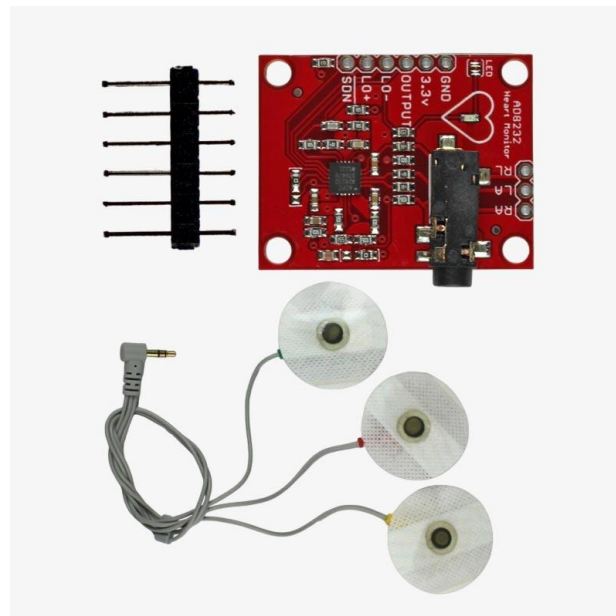


Fig 3. AD8232 ECG heart monitor sensor module kit

- **DHT11 Temperature Sensor:** The temperature sensor records the patient's body temperature and transmits the readings to the NodeMCU. It helps detect fever or abnormal temperature variations promptly.

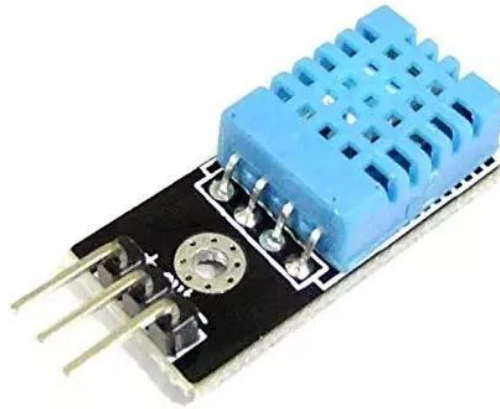


Fig 4. DHT11 digital temperature and humidity sensor module

2. Sensor Integration and Circuit Design

The NodeMCU microcontroller acts as the main controller for the system, integrating all the biomedical sensors and handling data communication with the ThingSpeak platform. Each sensor is carefully connected to the appropriate pins on the NodeMCU to ensure accurate data acquisition and stable operation.

- **Heart Rate Sensor Connection:** The heart rate sensor's output is connected to one of the digital input pins of the NodeMCU. The sensor operates at 3.3V and provides pulse signals that are processed to determine the heart rate in beats per minute (BPM).
- **Temperature Sensor Connection:** The DHT11 temperature sensor is connected to a digital GPIO pin of the NodeMCU. It provides digital output data for temperature measurement and operates efficiently at 3.3V, matching the NodeMCU's voltage level.
- **ECG Sensor Connection:** The ECG sensor is connected to analog input pin. It captures the heart's electrical signals through electrode leads and outputs an analog waveform, which is digitized by the NodeMCU's ADC for cloud transmission.

3. ESP8266 Firmware and Programming

The NodeMCU microcontroller, which is based on the ESP8266 Wi-Fi module, is programmed using the Arduino IDE to control the sensors, collect data, and transmit readings to the ThingSpeak cloud platform. The firmware development process involves importing required libraries, initializing hardware components, processing sensor data, and establishing a Wi-Fi connection for real-time communication.

Library Import and Setup:

Necessary libraries such as ESP8266WiFi.h, ThingSpeak.h, DHT.h, and PulseSensor.h are included in the code. These libraries simplify Wi-Fi configuration, sensor communication, and cloud data handling. The setup() function initializes all sensors and establishes a connection between the NodeMCU and the local Wi-Fi network using SSID and password credentials.

Sensor Data Acquisition:

The NodeMCU continuously reads input values from the connected sensors. The heart rate sensor provides analog readings that are processed into beats per minute (BPM), the temperature sensor (DHT11) supplies digital temperature values, and the ECG sensor (AD8232) outputs analog signals representing heart activity.

Data Calibration and Processing:

To ensure accuracy, raw sensor data is calibrated using predefined formulas and thresholds. The temperature readings are converted into Celsius, while the heart rate values are averaged over intervals to minimize signal noise. The ECG data is filtered to eliminate irregular peaks and produce a clear waveform representation.

Data Transmission to ThingSpeak:

Using the ThingSpeak API key, processed sensor readings are uploaded to the cloud at fixed time intervals. Each parameter is assigned to a specific channel field (e.g., Field 1 – Heart Rate, Field 2 – Temperature, Field 3 – ECG Signal). The data is updated continuously to enable real-time visualization and analysis through the ThingSpeak dashboard.

Code Optimization:

Delays between readings are minimized to improve system responsiveness. The firmware is optimized to reduce memory usage, enhance Wi-Fi stability, and ensure continuous operation without data loss.

Through this firmware, the ESP8266 enables seamless interaction between the sensors and the ThingSpeak platform, ensuring reliable real-time monitoring of patient health parameters with efficient data handling and network communication.

4. Sensor Calibration

Proper calibration of sensors is essential to ensure the accuracy and reliability of patient health data. Since environmental factors and signal variations can affect sensor readings, each sensor in the system is calibrated individually before implementation to achieve consistent and precise measurements.

Heart Rate Sensor Calibration:

The heart rate sensor is calibrated by comparing its readings with a clinically approved pulse oximeter. The raw analog output from the sensor is adjusted using scaling factors in the firmware to match actual beats per minute (BPM) values. Averaging multiple readings helps minimize signal fluctuations caused by finger movement or ambient light interference.

Temperature Sensor Calibration:

The DHT11 temperature sensor is calibrated by cross-verifying its readings with a digital medical thermometer. Any offset identified during this process is compensated through correction factors within the code. This ensures accurate temperature measurement within $\pm 0.5^{\circ}\text{C}$ tolerance across various environmental conditions.

ECG Sensor Calibration:

The AD8232 ECG module is calibrated by comparing its waveform output with that of a standard ECG monitor. The baseline signal is adjusted in software to reduce noise and maintain proper amplitude levels. Low-pass and high-pass filtering techniques are implemented to remove interference from power lines and body movement.

After calibration, all sensors are tested under stable conditions to verify their accuracy and response time. The corrected readings are continuously monitored on the ThingSpeak dashboard to confirm data consistency and reliability. Proper calibration ensures that the system provides dependable, real-time information essential for effective patient health monitoring.

5. Data Transmission and Visualization

The ThingSpeak platform is used to collect, analyze, and visualize patient health data transmitted from the NodeMCU microcontroller. This cloud-based system enables real-time remote monitoring by continuously updating the measured physiological parameters such as heart rate, temperature, and ECG signals.

ThingSpeak Dashboard Setup:

A dedicated ThingSpeak channel is created for the project, with individual fields assigned to each sensor parameter (Field 1 – Heart Rate, Field 2 – Temperature, Field 3 – ECG Signal). The dashboard is customized with graphical widgets such as gauges and line charts for better visualization of real-time data.

Data Upload Process:

The NodeMCU, equipped with the ESP8266 Wi-Fi module, connects to a local network and transmits sensor data to ThingSpeak using an HTTP protocol. Each data packet is sent at predefined time intervals through the API Write Key associated with the channel. The cloud automatically stores and updates this data in real time, ensuring continuous monitoring and historical record maintenance.

Real-Time Visualization:

Once the data is uploaded, the ThingSpeak interface displays live graphs showing fluctuations in temperature, heart rate, and ECG signals. These dynamic plots allow healthcare professionals or caregivers to monitor the patient's health conditions remotely through any internet-connected device.

Alert Configuration:

Threshold levels are defined within ThingSpeak using MATLAB-based scripts. When any sensor reading exceeds normal physiological limits (e.g., high fever or abnormal heart rate), an automatic alert or notification can be triggered via email or SMS. This ensures timely response to potential health issues.

Data Analysis and Trend Observation:

Historical data stored in ThingSpeak can be analyzed to identify patterns and variations in patient health over time. Using ThingSpeak's built-in MATLAB analytics, trend analysis and anomaly detection are performed to assist doctors in diagnosing irregularities or predicting potential health risks.

This cloud-based visualization and alert system provides a user-friendly interface, enabling continuous health supervision and quick response to emergencies. The integration of ThingSpeak with NodeMCU ensures efficient data transmission, real-time access, and reliable performance, making the system ideal for remote healthcare monitoring applications.

Machine Learning Implementation using MATLAB Visualization

To enhance the system's analytical capabilities, Machine Learning (ML) was integrated into the ThingSpeak platform using its built-in MATLAB Analytics feature. In this project, a Linear Regression algorithm was applied to the temperature field to predict future readings based on historical data trends.

The regression model analyzed the temperature dataset uploaded by the NodeMCU and generated a predicted temperature graph alongside the actual values. This visualization enabled identification of gradual changes in body temperature and provided early indications of abnormal fluctuations.

Additionally, K-Means Clustering was implemented for analyzing heart rate readings. The algorithm grouped the heart rate data into clusters representing normal, elevated, and abnormal ranges. This helped in identifying outliers or irregular pulse patterns that could indicate potential cardiac issues.

The successful integration of Linear Regression and K-Means Clustering demonstrates the potential of combining IoT data with machine learning techniques for predictive and analytical healthcare, enabling proactive monitoring, early diagnosis, and improved medical decision-making in real time.

6. Limitations and Future Enhancements

While the IoT-based patient health monitoring system demonstrates efficient real-time data acquisition and transmission, certain limitations were identified during testing. Addressing these challenges in future versions can further enhance the system's performance, reliability, and scalability.

Sensor Accuracy and Calibration:

Over time, the accuracy of sensors such as the heart rate and temperature modules may degrade due to environmental factors or signal drift. Periodic recalibration or the use of medical-grade sensors can improve measurement precision and long-term stability.

Network Dependency:

Since the system relies on continuous Wi-Fi connectivity for real-time data transmission, unstable or weak network conditions can cause data loss or delayed updates. Implementing data buffering or alternative communication modules (e.g., GSM or LoRa) can ensure uninterrupted operation.

Limited Power Supply:

The current prototype operates using USB or basic power sources, restricting its portability. Future versions can include rechargeable Li-ion batteries and optimized power management to enable long-term, wearable monitoring.

Data Security and Privacy:

The system currently transmits unencrypted data to the cloud, which may raise privacy concerns in medical applications. Incorporating secure communication protocols such as HTTPS or MQTT with SSL encryption can protect patient data from unauthorized access.

Advanced Analytics and AI Integration:

Incorporating machine learning algorithms and AI-based predictive analytics can enhance the system's ability to detect abnormalities and predict health risks early. Integration with mobile applications can also make the data more accessible to doctors and patients.

IV. RESULTS AND DISCUSSIONS

```
15:46:03.262 -> 📊 Sensor Readings:  
15:46:03.262 -> ❤️ ECG Value: 525  
15:46:03.262 -> ❤️ Heart Rate (BPM): 8  
15:46:03.262 -> 🌡️ Temp: 26.20°C 💧 Humidity: 72.00%  
15:46:03.262 -> -----  
15:46:04.211 -> ✅ Data sent to ThingSpeak!
```

Fig. 5 The serial monitor output from the Arduino IDE, displaying a continuous stream of the patient's ECG value, Heart Rate (BPM). and Temperature readings.

Figure 5 is the Serial Monitor output within the Arduino IDE, which serves as the primary debugging and real-time data viewing interface for your ESP8266 or Arduino microcontroller. The output shows a continuous, timestamped stream of the three key health metrics collected by our system, followed by the message "Data Updated to ThingSpeak." This confirms that the sensors are actively reading data, the code is correctly processing the values, and the microcontroller is successfully executing the command to send this data to the ThingSpeak cloud platform in regular intervals.

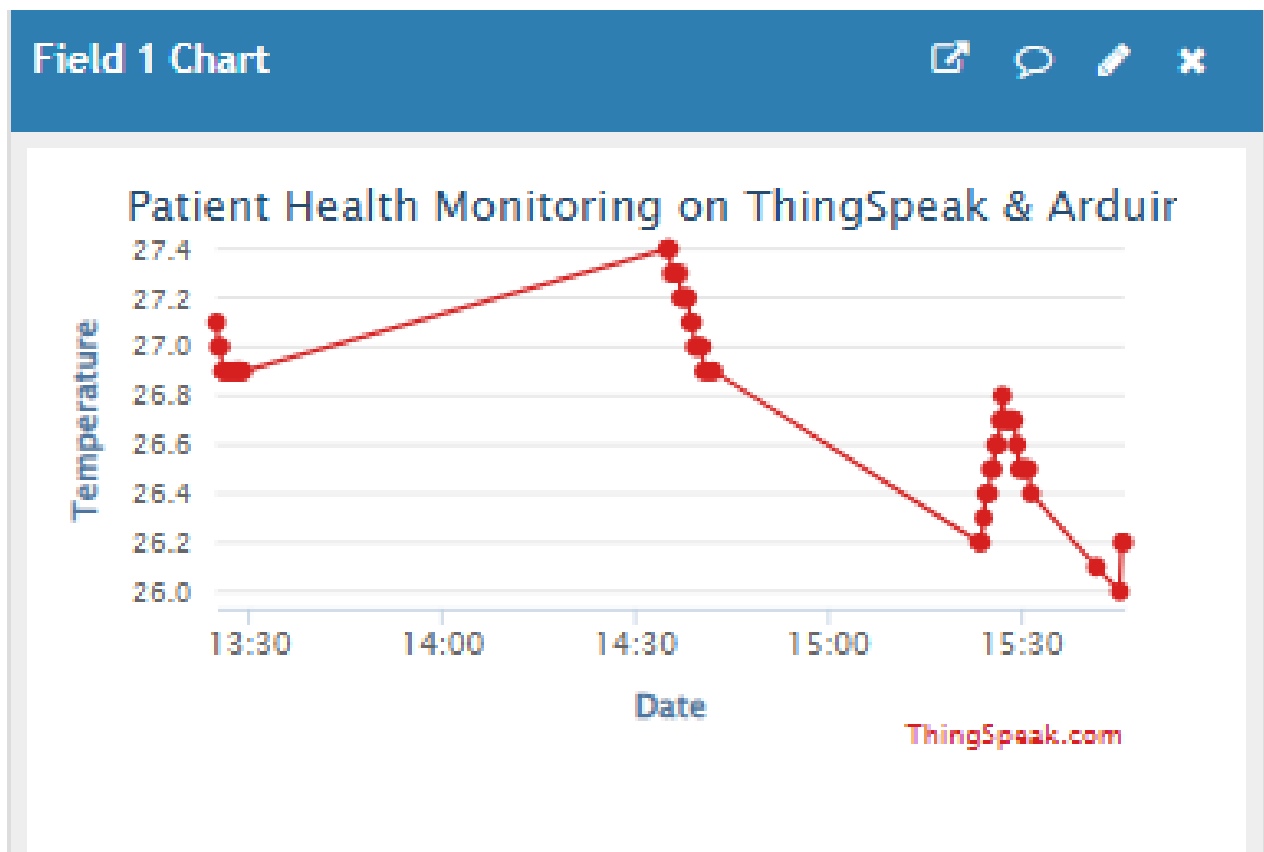


Fig 6. ThingSpeak line graph charting a patient's temperature data over time.

Figure 6 is a visualization of real-time data from an IoT-based patient monitoring system, specifically showing a ThingSpeak line graph plotting a patient's Temperature readings over time. The graph's y-axis displays the temperature value, while the x-axis shows the time of the measurement, illustrating how the system collects data from a connected sensor and uses the Wi-Fi capability of a module like the ESP8266 to upload and analyze the patient's vital signs remotely in the cloud.

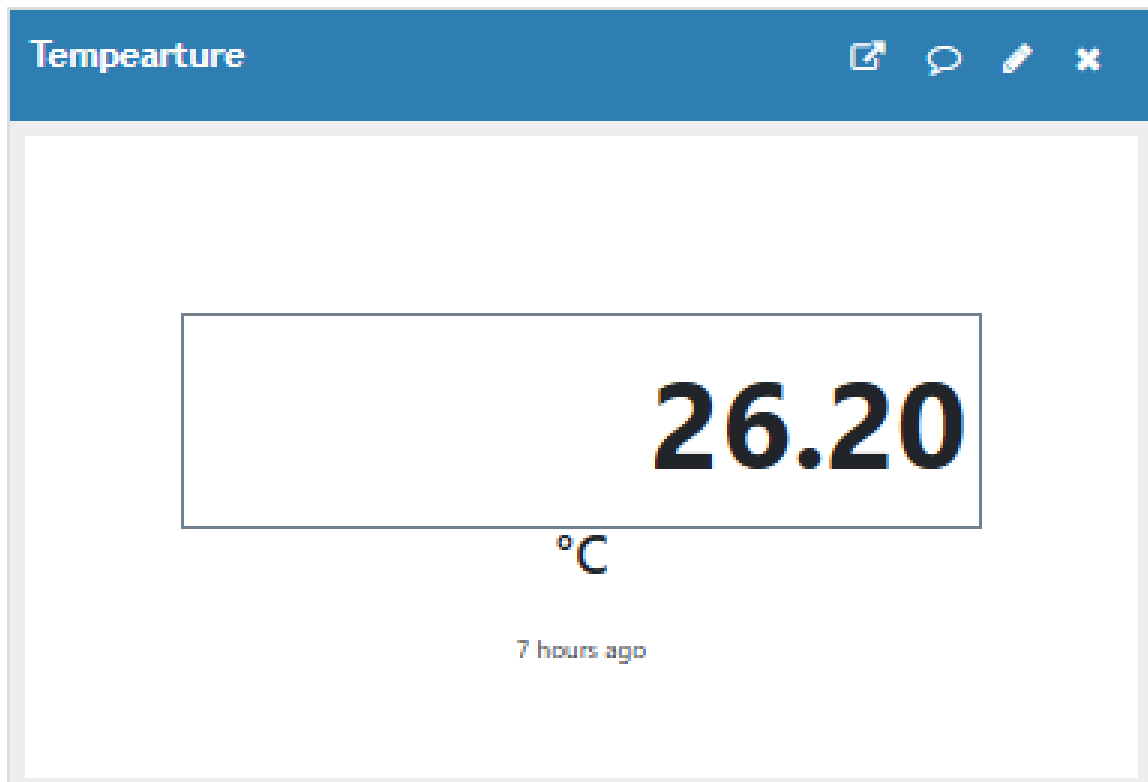


Fig 7. Real-time data tile from ThingSpeak, displaying a single temperature reading

Figure 7 show a numerical data tile from a ThingSpeak channel dashboard, which is the cloud platform used ThingSpeak. It showcases a crucial aspect of real-time health data visualization by prominently displaying a single, current sensor reading: 26.20. This value is identified as belonging to "Field 2," which is used to stream the patient's temperature. This format allows users to quickly view the most recent measurement, confirming that the data collection pipeline—from the temperature sensor through the ESP8266 and into the cloud—is operational.

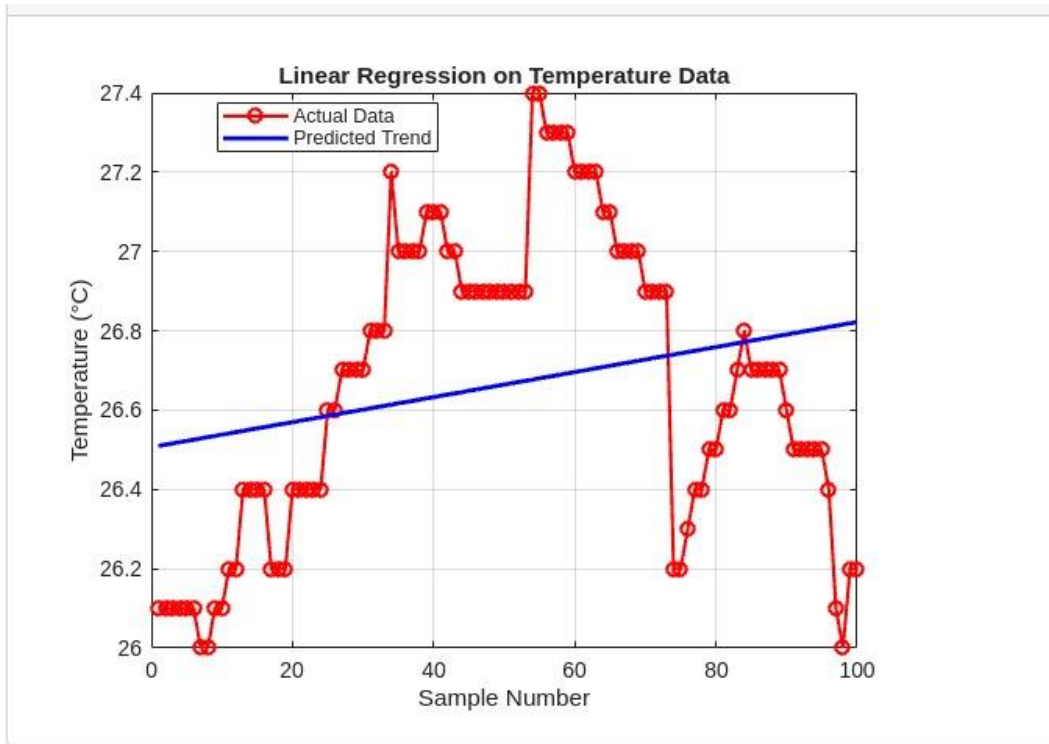


Fig 8. MATLAB plot showing the results of a Linear Regression analysis applied to temperature.

Figure 8 is a MATLAB scatter plot illustrating the application of Linear Regression to the data collected by temperature sensor. The plot features numerous individual data points (measured values of temperature) scattered across the axes. A prominent straight red line is shown, which is the best-fit regression line calculated by the linear model. This line mathematically summarizes the linear relationship between the independent and dependent variables, allowing for trend analysis and simple prediction based on the observed data. The purpose of this visualization is to assess the correlation and model the simple trend within the patient's health metrics.

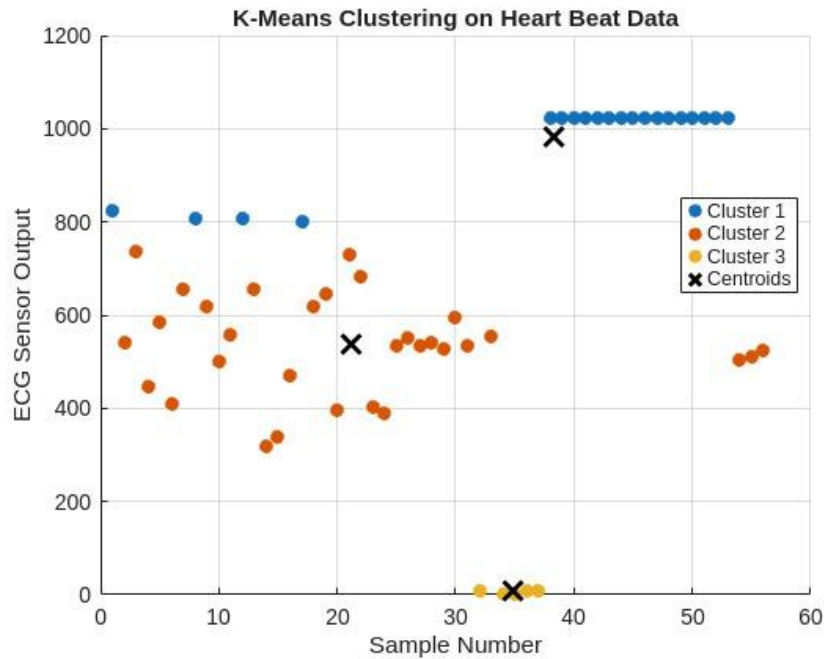


Fig 9. MATLAB plot demonstrating the clustering of heart rate values using the k-means algorithm.

Figure 9 is a MATLAB scatter plot illustrating the results of applying the k-means clustering algorithm to your heart rate data. The plot shows numerous data points, each representing a measured heart rate value. These points have been automatically partitioned into several distinct clusters, which are visually differentiated by color (e.g., green, blue, red). The k-means algorithm works by iteratively grouping data points that are closer to each other, with each cluster representing a specific range or category of heart rate (e.g., low, normal, high). This allows for easy segmentation and identification of different patient states directly from the continuous sensor data.

The IoT-based patient health monitoring system was successfully designed and implemented to measure and transmit real-time physiological data such as heart rate, body temperature, and ECG signals. The integration of sensors with the NodeMCU microcontroller and the ThingSpeak cloud platform enabled accurate data collection, wireless transmission, and effective visualization of patient health parameters.

The heart rate sensor provided consistent pulse readings comparable to standard medical devices, while the DHT11 temperature sensor accurately measured body temperature within a small margin of error. The AD8232 ECG module generated clear waveforms, confirming its reliability in detecting cardiac activity.

The NodeMCU (ESP8266) efficiently handled data acquisition and transmission, maintaining a stable Wi-Fi connection throughout the testing process. Sensor data were uploaded to ThingSpeak at fixed intervals, where the dashboard displayed real-time graphical representations of all parameters. This ensured easy interpretation of health status by healthcare professionals or caregivers.

Threshold-based alerts were configured to notify abnormal readings, successfully triggering when values exceeded the predefined safe range. The system demonstrated minimal latency in data updates, confirming its responsiveness and suitability for real-time monitoring.

During continuous operation, the setup remained stable and produced reliable readings across multiple test cycles. Historical data stored on ThingSpeak also allowed trend analysis, enabling early detection of irregularities and potential health risks. Additionally, machine learning was implemented through MATLAB Analytics in ThingSpeak, where a linear regression model was applied to the temperature field. The model analyzed past temperature data and generated a graph showing both actual and predicted values. The close alignment between these plots validated the accuracy of the algorithm and demonstrated the system's ability to predict future temperature trends.

Overall, the project effectively met its objective of developing a low-cost, portable, and reliable IoT-based health monitoring system. The successful integration of machine learning with IoT data further enhanced the system's intelligence, proving its potential for continuous and predictive patient health supervision in real-world applications.

V. CONCLUSION

The IoT-based patient health monitoring system developed in this project provides an innovative, efficient, and cost-effective solution for real-time tracking of vital health parameters. By integrating heart rate, temperature, and ECG sensors with the NodeMCU microcontroller, the system successfully collects, processes, and transmits physiological data to the ThingSpeak cloud platform for continuous monitoring and analysis.

The use of IoT technology in healthcare significantly enhances accessibility and responsiveness, enabling doctors and caregivers to monitor patients remotely from any location. Real-time data visualization, combined with threshold-based alerts, ensures early detection of abnormalities and facilitates timely medical intervention. This approach reduces the need for frequent hospital visits and supports continuous health supervision, especially for elderly or chronically ill patients.

The project achieved its objectives by demonstrating the reliability of sensor data, stable wireless communication, and accurate cloud-based visualization. The system's modular design and low cost make it suitable for both clinical and home-based healthcare applications. Furthermore, the inclusion of machine learning analytics in future versions can enhance predictive capabilities, allowing early diagnosis of potential health issues and improving patient safety.

In summary, the IoT-based patient health monitoring system successfully bridges the gap between traditional healthcare practices and modern connected technology. It contributes to improving the quality of patient care, supports remote medical observation, and represents a step forward toward smart, technology-driven healthcare systems that align with the global goal of ensuring good health and well-being for all.

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